

صل علي سيدنا محمد
ادعوا لي بالتوفيق ولكم بالمثل
افتكروا ابي و جدتي في دعائكم

Lecture 5

Web Testing Issues

- Testing a web application is quite a bit more difficult than testing the same functionality in a Windows Desktop application
- Various issues are present:
 - Timing difficulties
 - Differences between browsers
 - New technologies that make web browsing a rich, but difficult-to-test, experience

Web Engineering Project Metrics

Example Metrics:

1. Number of **static Web** pages
2. Number of **dynamic Web** pages
3. Number of **internal** page **links**
4. Number of persistent **data objects**
5. Number of **external** systems interfaced
6. Number of executable **functions**

Web Engineering Project Metrics (2)

- Let,
 - N_{sp} = number of static Web pages
 - N_{dp} = number of dynamic Web pages
 - Then,
 - **Customization index**, $C = N_{dp} / (N_{dp} + N_{sp})$
 - The value of C ranges from 0 to 1
- 
- static $\leftarrow$ $\rightarrow$ dynamic
- 0.5 = half dynamic

Example Usability Measures

6.1- Success score

Success, or completion means that a user managed to complete a task that they were given. The basic formula for the success score is:

$$\text{Success Score} = \frac{\text{number of successful tasks} \quad \text{Nº OF COMPLETED TASKS}}{\text{TOTAL Nº OF ATTEMPTS} \quad \text{total number of tasks attempted}}$$

The higher the success rate the better the user experience.

Example Usability Measures

6.2- Number of errors

In user testing, an **error** is any wrong action performed while completing a task. There are two types of errors: slips and mistakes.

The goal is right, the execution is wrong.

Slips are those errors that are made with the **right goal** (for example, a typo when entering the date of birth), and **mistakes** are errors made with the **wrong goal** (for instance, entering today's date instead of birth date).

There are two ways of measuring errors: measuring all of them (**error rate**) or focusing on one error (**error occurrence rate**).

Mistakes: The goal itself is wrong — due to misunderstanding or unclear design.

Error Rate =

Nº OF ERRORS

TOTAL Nº OF ATTEMPTS

Error Occurrence Rate =

TOTAL Nº OF ERRORS

TOTAL Nº OF POSSIBLE ERRORS

Example Usability Measures

6.6- System Usability Scale (SUS)

For those who don't trust the single-question solution, there is a list of 10 questions, known as the System Usability Scale. Based on the answers, the product gets a score on a scale from 0 to 100 (each question is worth 10 points).

the average SUS is 68 points. Results over 80 are considered excellent.

System Usability Scale Questionnaire

	Strongly Disagree			Strongly Agree	
1. I think that I would like to use this product frequently.	1	2	3	4	5
2. I found the product unnecessarily complex.	1	2	3	4	5
3. I thought the product was easy to use.	1	2	3	4	5
4. I think that I would need the support of a technical person to be able to use this product.	1	2	3	4	5
5. I found the various functions in the product were well integrated.	1	2	3	4	5
5. I found the various functions in the product were well integrated.	1	2	3	4	5
6. I thought there was too much inconsistency in this product.	1	2	3	4	5
7. I imagine that most people would learn to use this product very quickly.	1	2	3	4	5
8. I found the product very awkward to use.	1	2	3	4	5
9. I felt very confident using the product.	1	2	3	4	5
10. I needed to learn a lot of things before I could get going with this product.	1	2	3	4	5

Test Automation

Test Automation is the use of special software (separate from the software being tested) to control the execution of tests and the comparison of actual outcomes with predicted outcomes.

What is Selenium?

- **SELENIUM** is a free (open-source) automated testing framework used to validate web applications across different browsers and platforms.
- Selenium Software is not just a single tool but a suite of software, each piece catering to different testing needs of an organization.
- Here is the list of tools:
 1. Selenium Integrated Development Environment (IDE)
 2. Selenium Remote Control (RC)
 3. WebDriver
 4. Selenium Grid

Selenium Features

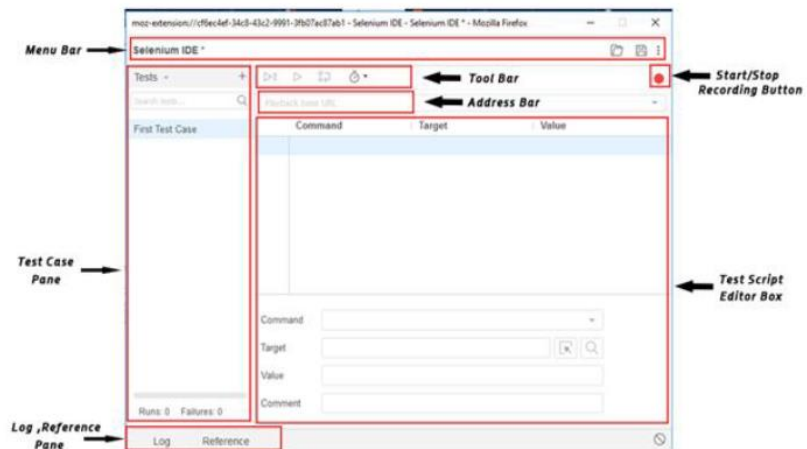
Windows, macOS, Linux).

1. Selenium is an open source and portable Web testing Framework.
2. Selenium IDE provides a playback and record feature for authoring tests without the need to learn a test scripting language. 🎧
3. It can be considered as the leading cloud-based testing platform which helps testers to record their actions and export them as a reusable script with a simple-to-understand and easy-to-use interface.
4. It also supports parallel test execution which reduces time and increases the efficiency of tests.
5. Selenium requires fewer resources as compared to other automation test tools.
6. Selenium web driver does not require server installation, test scripts interact directly with the browser.
7. Selenese is the language used to write Selenium Commands

Features Selenium IDE

- Selenium IDE is divided into different components, each having their own features and functionalities. We have categorized seven different components of Selenium IDE, which includes:

- Menu Bar
- Address Bar
- Tool Bar
- Test Case Pane
- Test Script Editor Box
- Start/Stop Recording Button
- Log, Reference Pane



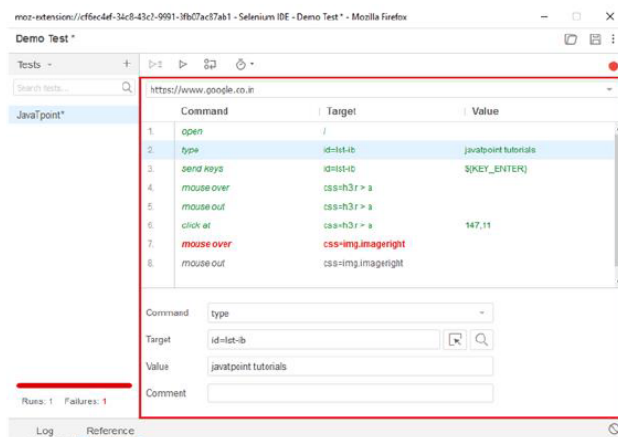
5. Test Script Editor Box

- Test Script Editor Box displays all of the test scripts and user interactions that were recorded by the IDE.
- Each user interaction is displayed in the same order in which they are performed.
- The Editor box is divided into three columns: Command, Target and Value.

1. Command – tells Selenium what to do (e.g. “open”, “type”, “click”, “verifyText”)

2. Target – tells Selenium which HTML element a command refers to (e.g. textbox, header, table)

3. Value – used for any command that might need a value of some kind (e.g. type something into a textbox)



Command Target Value

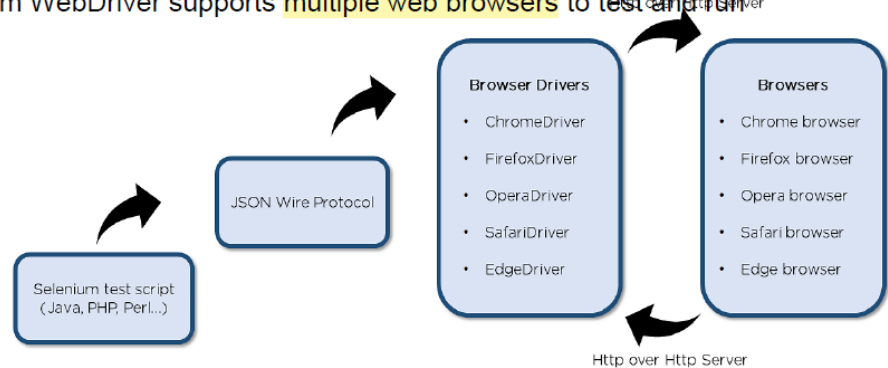
Common Commands (Actions)

Command	Description
<code>open (URL)</code>	This command opens specific URLs , and this command is used for both relative and absolute URLs .
<code>find_element(by, value)</code>	This command finds a single web element using different strategies.
<code>click ()</code>	This command allows clicking on web elements like links, buttons, checkboxes, or radio buttons.
<code>clickAt (locator, coord)</code>	This command allows you to click on an element with the help of pointers and coordinate strings .
<code>doubleClick ()</code>	This command allows a double click on the web element based on the specified content .
<code>focus ()</code>	This command moves the focus to a specified web element.
<code>refresh()</code>	This command allows clicking on the refresh button .
<code>pause(time)</code>	This command allows waiting for a specific amount of time in milliseconds .
<code>close()</code>	This command closes the current browser window that the WebDriver is using.
<code>mouseUp()</code>	Upon releasing the mouse button, this command reproduces the event that occurs.
<code>mouseDown()</code>	This command is used to perform left mouse click operation.
<code>mouseDownAt (locator, coord)</code>	Upon releasing the left mouse button at a particular location , this command reproduces the event that occurs.
<code>echo(statement)</code>	This command prints specified messages or statements on the console.

Selenium WebDriver Architecture

The 4 main Components of this architecture are

- **Selenium test script** - Selenium test script is the **test code written** in any of the mentioned **programming languages** that are interpreted by the driver
- **JSON Wire Protocol** - JSON Wire Protocol provides a transport mechanism to **transfer data between a server and a client**. JSON Wire Protocol is the industry standard for various web services
- **Browser drivers** - Selenium uses **drivers, specific to each browser** to establish a secure connection with the browser **حلقة الوصل مع المتصفح**
- **Browsers** - Selenium WebDriver supports **multiple web browsers to test and run** applications on.

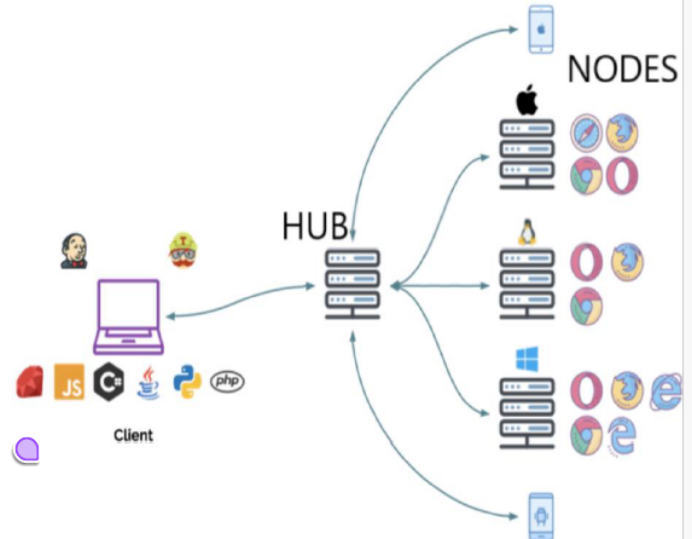


Pros & Cons of Selenium WebDriver

Pros	Cons
• Simpler installation than Selenium RC • Communicates directly to the browser • Browser interaction is more realistic • No need for a separate component such as the RC Server • Faster execution time than IDE and RC	• Installation is more complicated than Selenium IDE • Requires programming knowledge • Cannot readily support new browsers • Has no built-in mechanism for logging runtime messages and generating test results

Selenium Grid

- The two major components of Selenium Grid are:
 - Hub** is a **server** that accepts the access requests from the WebDriver client, routing the JSON test commands to the **remote drives on nodes**. It takes instructions from the client and executes them remotely on the various nodes in parallel
 - Node** is a remote device that consists of a native **OS** and a **remote WebDriver**. It receives requests from the hub in the form of JSON test commands and executes them using **WebDriver**



صل علي سيدنا محمد

ادعوا لي بالتوفيق ولكم بالمثل

افتكروا ابي و جدتي في دعائكم

Lecture 6

A1- Injection

- Finding a way to send text to a web application or browser that is interpreted as a command or code
- Tricks systems or browsers in to taking action
- Attacker sends simple text-based attacks that exploit the syntax of the targeted interpreter. Almost any source of data can be an injection vector, including internal sources.

SQL injection prevention techniques

Primary Defenses: 🌀

- Option 1: Use of Prepared Statements (with Parameterized Queries)
- Option 2: Use of Properly Constructed Stored Procedures
- Option 3: Allow-list Input Validation
- Option 4: Escaping All User Supplied Input 🌀

Additional Defenses:

- Also: Enforcing Least Privilege
- Also: Performing Allow-list Input Validation as a Secondary Defense
- Always perform 'white list' input validation on all user supplied input

Parameterized Statements

Use of Prepared Statements (with Parameterized Queries)

Programming languages talk to SQL databases using **database drivers**. A driver allows an application to **construct and run SQL statements against a database**, extracting and manipulating data as needed.

Parameterized statements make sure that the parameters (i.e. inputs) passed into SQL statements are **treated in a safe manner**.

The user **input values** are **passed as parameters**. They can **no longer contain an executable code** since the **parameter is treated as a literal value** **تعامل كقيم وليست أوامر** and **checked for the type and length**.

What are cookies and sessions?

A cookie is a small piece of data from a web site that is stored on a visitor's browser to help the website track the **visitor's activity** on the web site. Sessions and cookies are sometimes conflated, creating confusion. More specifically, session IDs and cookie IDs are confused.

While they are closely related, they **are not the same thing**. A cookie identifies, often anonymously, a specific visitor or a specific computer. **Cookies can be used for authentication, storing site preferences, saving shopping carts, and server session identification**

What are cookies and sessions?- 2

Server logs typically contain both the session ID and cookie ID of a visitor.

- A web session ID is unique to a specific visit, 
- while a cookie id is unique to a specific visitor and thus (developers hope) remains the same through multiple web sessions. 

By mapping a single cookie ID to multiple session IDs, developers and analysts can get a clearer picture of how visitors interact with their web applications.

session management types



There are two main types of web session management: client-side and server-side.

1. **Client-side session management** stores the session data on the client's browser. This is typically done using cookies, which are small pieces of data that are stored on the client's computer. When the client makes a request to the server, the cookie is sent with the request. The server can then use the cookie to identify the client and retrieve the associated session data.
2. **Server-side session management** stores the session data on the server. When a client connects to the server, the server generates a unique session identifier and sends it to the client. The client stores the session identifier and sends it back to the server with each request. The server then uses the session identifier to retrieve the associated session data.

1-Client-side session management methods

- **Cookies:** Cookies are the most common type of client-side session management. They are simple to implement and use, and they are supported by all major browsers. However, cookies can be disabled by users, and they can be stolen by attackers. 🌐
- **Hidden form fields:** Hidden form fields are another way to store session data on the client side. When a client submits a form, the hidden form fields are sent to the server with the request. The server can then use the hidden form fields to identify the client and retrieve the associated session data. However, hidden form fields can be easily tampered with by attackers.
- **Browser storage:** Browser storage mechanisms such as local storage and session storage can also be used to store session data on the client side. Browser storage is more secure than cookies because it cannot be accessed by other websites. However, browser storage is not supported by all browsers.

صل علي سيدنا محمد

ادعوا لي بالتوفيق ولكم بالمثل

افتكروا ابي و جدتي في دعائكم

2- Server-side session management methods

- **URL rewriting:** URL rewriting is a technique that can be used to store the session identifier in the URL. When a client makes a request, the server rewrites the URL to include the session identifier. The client then sends the rewritten URL to the server with each request. However, URL rewriting can be cumbersome to implement, and it can be vulnerable to attacks.
 - **HTTP headers:** HTTP headers can also be used to store the session identifier. When a client makes a request, the server sends the session identifier in an HTTP header. The client then sends the HTTP header to the server with each request. However, HTTP headers can be tampered with by attackers.
 - **In-memory storage:** In-memory storage is a simple way to store session data on the server. The server stores the session data in memory and retrieves it when needed. However, in-memory storage is not scalable, and it can be lost if the server crashes.
 - **Database storage:** Database storage is a more scalable and reliable way to store session data on the server. The server stores the session data in a database and retrieves it when needed. However, database storage can be more complex to implement than other server-side session management methods.
-
- Use **server-side session management** for applications that handle sensitive data, such as financial or medical information.
 - Use **client-side session** management for applications that do not handle sensitive data and where performance is a critical concern.
 - Use a **combination** of client-side and server-side session management for applications that require both performance and security.

Avoiding Broken Authentication and Session Management

- Verify your architecture
 - **Authentication** should be simple, centralized, and standardized
 - Use the standard session id provided by your container
- Verify the implementation
 - **Test** all the authentication-related functions
 - Verify that **logout** actually destroys the session

A3 – Cross-Site Scripting (XSS)

- Attacker sends **text-based attack scripts** that exploit the interpreter in the browser. Almost any source of data can be an attack vector, including internal sources such as data from the database.
- Cross-site scripting attacks are those in which attackers inject malicious code **كود ضار**, usually client-side scripts, into web applications.
- Because of the number of possible injection locations and techniques, many applications are vulnerable **معرضة لخطر** to this attack method.

صل علي سيدنا محمد

ادعوا لي بالتوفيق ولكم بالمثل

افتكروا ابي و جدتي في دعائكم

Cross Site Scripting (XSS)- types

1. NonPersistent

it requires a user to visit the specially crafted link by the attacker. When the user visit the link, the crafted code will get executed by the user's browser.

2. Persistent

it occur in either community content driven websites or Web mail sites and do not require specially crafted links for execution.

persistent XSS much more dangerous than non-persistent because the user has no means of defending himself. Once a hacker has his exploit code in place, he'll again advertise the URL to the infected

Safe ways to represent dangerous characters in a web page

Characters	Decimal	Hexadecimal	HTML Entity	Unicode
" (double quotation marks)	"	"	"	\u0022
' (single quotation mark)	'	'	'	\u0027
& (ampersand)	&	&	&	\u0026
< (less than)	<	<	<	\u003c
> (greater than)	>	>	>	\u003e

Definition of an origin

(https://developer.mozilla.org/en-US/docs/Web/Security/Same-origin_policy)

- Two pages have the **same origin** if the following are the same in both pages:
 - **Protocol** http or https
 - **port (if one is specified),** :8080
 - **and host** name of website name.com
- The following table gives examples of origin comparisons to the URL **http://store.company.com/dir/page.html**:

URL	Outcome	Reason
http://store.company.com/dir2/other.html	Success	
http://store.company.com/dir/inner/another.html	Success	
https://store.company.com/secure.html	Failure	Different protocol
http://store.company.com:81/dir/etc.html	Failure	Different port
http://news.company.com/dir/other.html	Failure	Different host

صل علي سيدنا محمد

ادعوا لي بالتوفيق ولكم بالمثل

افتكروا ابي و جدتي في دعائكم

صل علي سيدنا محمد
ادعوا لي بالتوفيق ولكم بالمثل
افتكروا ابي و جدتي في دعائكم

Lecture 8

Performance Testing Metrics: Parameters Monitored - 1

- **Processor Usage** – an amount of time processor spends executing non-idle threads.
- **Memory use** – amount of physical memory available to processes on a computer.
- **Disk time** – amount of time disk is busy executing a read or write request.
- **Bandwidth** – shows the bits per second used by a network interface.
- **Private bytes** – number of bytes a process has allocated that can't be shared amongst other processes. These are used to measure memory leaks and usage.
- **Committed memory** – amount of virtual memory used.
- **Network bytes total per second** – rate which bytes are sent and received on the interface including framing characters.
- **Response time** – time from when a user enters a request until the first character of the response is received.

Performance Testing Metrics: Parameters Monitored - 3

- **Throughput** – rate a computer or network receives requests per second.
- **Amount of connection pooling** – the number of user requests that are met by pooled connections . قنوات سابقة التحضير . The more requests met by connections in the pool, the better the performance will be.
- **Maximum active sessions** – the maximum number of sessions that can be active at once.
- **Hit ratios** – This has to do with the number of SQL statements that are handled by cached data instead of expensive I/O operations. This is a good place to start for solving bottlenecking issues.
- **Hits per second** – the no. of hits on a web server during each second of a load test.

Response Time and Latency

- **Response time** = time for an operation to complete, including the time spent waiting and being serviced
- **Latency** = time an operation spent waiting in a queue
- **Service time** = Actual time to perform job continuously.
- **Response time** = Latency + Service time

Latency - HTTP GET Request

- Assume that a GET request consists of:
 - DNS resolution → (Latency)
 - Establishing TCP connection → (Latency)
 - Data transfer from the server to the client (Service Time)

Throughput

- Throughput is a measure of how many units of information a system can process in a given amount of time (Second).
-
- The rate at which work is completed
- The meaning depends on the target evaluated
- Example measures:
 - Bytes / bits per second
 - SQL queries per second
 - REST calls per second
 - Transactions per second

IOPS

- Input / output operations completed per second
- Throughput-oriented metric
- The meaning depends on the target evaluated
- Examples:
 - Network devices (TCP) – packets received / sent per second
 - Block devices – reads / writes per second

Utilization

There are 2 definitions:

1. **Time-based definition:** How much time a resource is busy during a period of time.
 - CPU Utilization
 - Disk Utilization
 - 100% Utilization doesn't have to lead to the saturation of the resource. We need to examine other metrics of the resource that indicates its saturation
2. **Capacity-based definition:** The extent to which the capacity مقدار استغلال القدرات المتاحة of a resource was used during a time period.

Note about capacity utilization

- it assumes that every resource (component) has some maximum capacity it can deliver.
- However, it's very **difficult to determine** the maximum capacity for some resources,
- for instance hard disks. To determine the maximum capacity of hard disks, the other parts of computer systems that disks use must be evaluated.

Saturation

- The extent to which a resource has **queued** work because it **cannot accept** more work التعطل بسبب الوصول للذروة
- Capacity-based Utilization $\geq 100\%$
- **Increases** latency and response time
- **Bottleneck** = the **resource** that **limits** performance

Scalability

- How does the performance of the product **change** under **increasing load**?
- **Load** meanings examples:
 - Hourly number of users accessing the product
 - API calls per second
 - Amount of data uploaded to the product per second
- System may perform in normal load but may **not be** scalable

Stability 🍌

- Does the system perform over **long time**?
- Does performance gets back to normal after an exceptional situation?
- **Memory leaks** (objects created and no more required, but still in memory)
- Excessive logging on disk devices

More Types of Performance Testing

- **Spike testing** – tests the software's reaction to **sudden large spikes** ارتفاع مفاجئ في الأحمال in the load generated by users.
- **Volume testing** – Under Volume Testing **large number of Data** is populated in a database, and the overall software **system's behavior is monitored**. The objective is to **check** software application's **performance** under varying database volumes.

Jmeter Overview

- Open source tool for Load and Performance testing
- Java desktop application
- Runs on any platform supporting Java (Windows, Linux, Mac)
- Simulates multiple concurrent users (Multi-threading) to create heavy load

Load & Stress Testing

• JMeter Performance Testing includes:

1. **Load Testing:** Modeling the **expected usage** by simulating multiple user access the Web services concurrently. Simulating expected usage (multiple concurrent users).
2. **Stress Testing:** Every web server has a **maximum load capacity**. When the load **goes beyond** تجاوز الحد الأقصى the limit, the web server starts responding **slowly and produce errors**. The purpose of the Stress Testing is to **find the maximum load** the web server can handle.

Going beyond the limit to find the **maximum load capacity**.

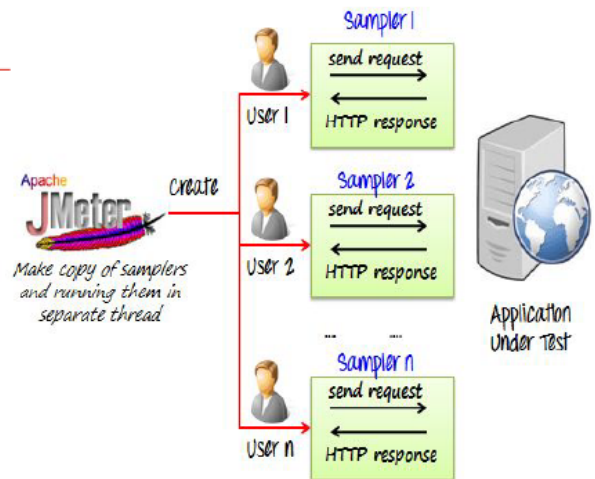
JMeter Features

- **Open source license:** JMeter is totally **free**, allows developer use the source code for the development
- **Friendly GUI:** JMeter is extremely easy to use and doesn't take time to get familiar with it
- **Platform independent:** JMeter is 100% pure Java desktop application. So it can **run on multiple platforms**
- **Full multithreading framework.** JMeter allows **concurrent and simultaneous sampling** of different functions by a separate thread group
- **Visualize Test Result:** Test result can be displayed in a different format such as **chart, table, tree** and log file
- **Easy installation:** You just copy and run the *.bat file to run JMeter. No installation needed.
- **Highly Extensible:** You can write your own tests. JMeter also supports visualization plugins allow you to extend your testing



JMeter Features (2)

- **Multiple testing strategy:** JMeter supports many testing strategies such as **Load Testing**, **Distributed Testing**, and **Functional Testing**.
- **Simulation:** JMeter can simulate **multiple users** with **concurrent threads**, create a **heavy load** against web application under test
- **Support multi-protocol:** JMeter does not only support web application testing but also evaluate database server performance. All basic protocols such as HTTP, JDBC, LDAP, SOAP, JMS, and FTP are supported by JMeter
- **Record & Playback - Record** the user activity on the browser and simulate them in a web application using JMeter
- **Script Test:** Jmeter can be **integrated** with other tools like Selenium for **automated testing**.



Jmeter Vs Load Runner



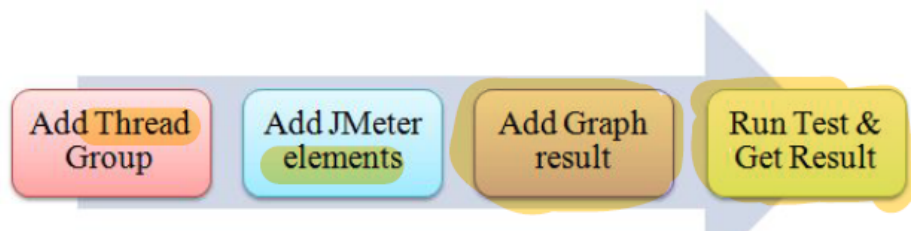
Jmeter	HP Load Runner
Open source and free to use	It is a licensed tool and costly
Jmeter is <u>cross platform</u>	It works with <u>Window</u> OS and <u>Linux</u>
It has <u>unlimited load</u> generation capacity	It has <u>limited load</u> generation capacity
It is <u>standalone</u> application	It has <u>3 separate</u> application
Scripting <u>isn't</u> essential in Jmeter	It requires <u>scripting knowledge</u>
Elements are <u>easier to define</u> in Jmeter	Configuring each element is <u>more complex</u> . They <u>all require complex scripting in C</u>

Traditional scenario for using JMeter

Before testing the performance of target web application, we should determine-

- **Normal Load:** Average number of users visit your website
- **Heavy Load:** The maximum number of users visit your website
- What is your **target** in this test?

Here is the roadmap of traditional scenario for using jmeter



2. Create test plan

A useful test plan is created with minimum 3 components –

- **Thread Group** –
 - contains the simulation of multiple concurrent users.
 - A single thread represent a single user.
 - We can create any number of threads to put the desired load on the application.
 - It also help us in scheduling the delay between two threads, and any repetition of request batches.
- **HTTP Request** –
 - consist the HTTP request configuration which thread group will be invoking.
 - It is the application URL which you want to load test.
- **Listener** –
 - helps in viewing the result of the whole testing process.
 - There are multiple listener available in jmeter to verify the testing results.

In this Example:

- **Number of Threads:** 100 (Number of users connects to the target website: 100)
- **Loop Count:** 10 (Number of time to execute testing)
- **Ramp-Up Period:** 100

Thread Group

Name: Thread Group

Comments:

Action to be taken after a Sampler error: ☒ Continue

Thread Properties

Number of Threads (users): 100

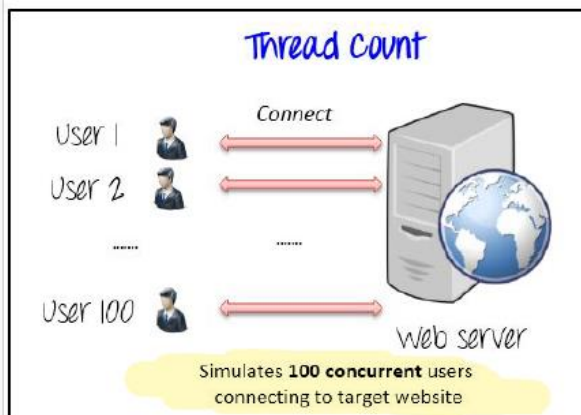
Ramp-Up Period (in seconds): 100

Loop Count: ☐ Forever 10

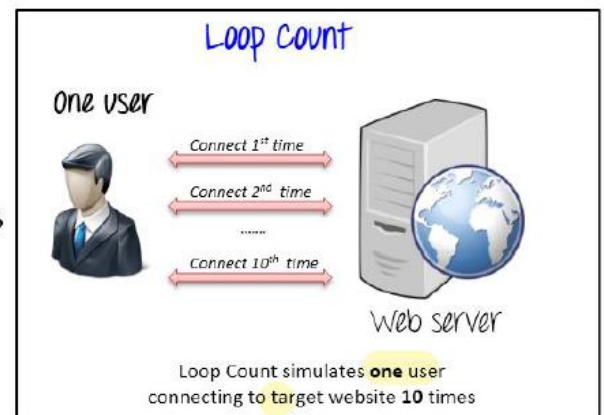
☐ Delay Thread creation until needed

☐ Scheduler

The Thread Count and The Loop Counts are **different**



VS

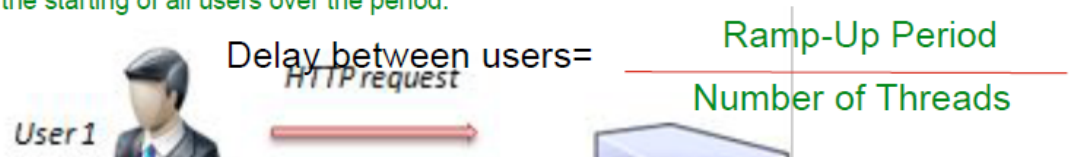


Total Requests = Thread Count * Loop Count

Ramp-Up Period

- Ramp-Up Period tells JMeter how long to **delay** before starting the next user. For example, if we have 100 users and a 100-second Ramp-Up period, then the delay between starting users would be 1 second (100 seconds / 100 users)

Distributes the starting of all users over the period.



,

Deviation Throughput Median

At the bottom of the picture, there are the following statistics, represented in colors:

- **Black:** The **total** number of **current samples sent**.
- **Blue:** The current **average** of all samples sent.
- **Red:** The current **standard deviation**. ↓
- **Green:** **Throughput rate** that represents the **number of requests per minute the server handled** ↗

Example of Results Analysis - 2

- To analyze the performance of the web server under test, you should focus on 2 parameters
 - a. **Throughput**
 - b. **Deviation**
- The **Throughput** is the most important parameter. It represents the ability of the server to handle a heavy load. The **higher the Throughput is, the better is the server performance**.
- In this test, the throughput of Google server is 1,491.193/minute. It means Google server can handle 1,491.193 requests per minute. This value is quite high so we can conclude that Google server has good performance
- The **deviation** is shown in red – it indicates the deviation from the average. The **smaller the better**.

صل علي سيدنا محمد
ادعوا لي بالتوفيق ولكم بالمثل
افتكروا ابي و جدتي في دعائكم

Lecture 9&10

NATIVE APPS-2

PROS	CONS
Easy low-level hardware access services.	Code Reusability : Low
Easy access to high level services important to personal mobile experience.	Development & maintenance: Time-consuming & expensive
Full use of all functionalities that modern mobile devices have to offer.	Designers are required to be familiar with different UI components of each OS.
High usability.	.Upgrade flexibility: Low

1-CROSS-COMPILATION

- Separates **build environment** from **target environment**.
- **Platform-independent API** using a mainstream programming language like JavaScript, Ruby or Java.
- The **cross-compiler** then **transforms** the **code** into **platform-specific native apps**.
- The software artifact **generated** can be **deployed** and **executed natively** on the device.

ADVANTAGES:

- **Improved performance** and **User Experience**.
- **Full access** to functionalities of **underlying mobile OS** and device **specific capabilities**.

DISADVANTAGES:

- **Highly complex as cross-compilers** are **difficult** to program.
- Need to be keep **API consistent with different mobile platforms** and operating systems available.

2-VIRTUAL MACHINE APPROACH

- A **virtual machine layer** is used to **abstract** the **target platform details** from the **application's running code**.
- The framework **provides** both the **API** and **runtime environment**.
- The **runtime** executes on the mobile device and enables **interoperability** between the **device's OS** and the mobile application.

ADVANTAGES:

- **Full access to functionalities of underlying mobile OS and device specific capabilities.** *like compiler*
- **Portability:** VM's are **easier to maintain** & more **flexible** to extend.

DISADVANTAGES:

- **Slower** due to **runtime interpretation latency**.

What is a virtual machine?-1

- A virtual machine (VM) is an application that functions as a **virtual computer** but resides **within** a **physical computer** (e.g., a **host computer**).
- Often referred to as **guest machines**, virtual machines work as **separate independent machines** but **run as a process** on the **host machine's operating system**.
- However, **both** the **physical computer** and the **virtual machine** **believe** they are **each interacting** with a **separate physical machine**.

3-MOBILE WEB APPS

- Use standard web technologies such as **HTML 5, CSS 3 & JavaScript**.
- Features of **HTML 5** :
 - **Advanced UI** components,
 - access to **rich media** types,
 - **geolocation** services
 - **offline** availability.
- **Increasing popularity of HTML 5** in rendering engines such as **WebKit**.
- Runs on a **standalone mobile web browser**.
- Installed **shortcut**, launched like a **native app**.
- **UI logic** resides **locally**; makes the app **responsive and accessible offline**.

ADVANTAGES:

- **Multiplatform** support.
- **Low** development cost.
- Leverage **existing knowledge of web technologies**.

DISADVANTAGES:

- **Limited access** to OS API's.

4-HYBRID Mobile APPS

- Combines **native development** with **web technology**.
- The **web app runs inside** a thin **wrapper native app** يعمل داخل تطبيق يحتويه.
- The **wrapper native app** uses the **OS API's** to **create** an **embedded HTML rendering engine** which provides a **bridge** between the browser and device API's.
- The communication between **web app** and **native app** normally happens over **JavaScript** via custom built API's.

ADVANTAGES:

- **Flexibility** of web apps combined with **feature richness** of native apps.
- **Simplified** deployment and **immediate availability**.
- Leverage **existing knowledge**.

DISADVANTAGES:

- **Poorer user experience** as compared to **native apps**.
- **Restricted Access** to **advanced device capabilities**.

CROSS-PLATFORM FRAMEWORKS

PROS مزایا
Code Reusability
Plugins <i>پلاگین</i>
Easy for web developers
Reduced development costs
Support for enterprise & cloud services
Easy Deployment

CONS عیوب
Might not support every feature of OS OS Feature Limits
Cannot use own tools/IDE
Slower.
High end graphics & 3D support limited
Vendor lock-in



Real Device and Emulators Comparison

Criteria	Real testing device	Virtual testing device
Cost	Buying real devices at scale is cost prohibitive عائق مادي	Minimal cost incurred as oftentimes one can install them for free
Reliability	Real devices exhibit accurate results and allow for testing in the same condition as a user	Virtual testing devices only mimic تحاكي the device and can't replicate real user conditions like hardware and software configurations
Processing Speed	Software testing on real devices is much faster	Software testing on virtual devices is slower due to Binary translation
Suitable for Debugging	Debugging with real testing devices could be tricky , especially while capturing defects	Virtual devices make step by step debugging easy with the features, where you can capture the defects
Cross-Platform Testing	Cross-Platform Testing can be conducted normally using real devices	✓ Cross-Platform Testing can be seamlessly conducted using virtual devices (works in a less degree than real devices)

	Appium	XCUITest	Espresso
Application Type	Mobile web, native/hybrid mobile apps	Native/hybrid mobile apps	Native/hybrid mobile apps
Operating Systems	Cross-platform	iOS	Android
Languages	Language agnostic	Objective-C/Swift	Java/Kotlin
Key Users	QA, Test engineers	Developers (iOS)	Developers (Android)
Security Testing Type	Black box	Gray box	Gray box
Source Code Required	No	Yes	Yes
Speed	Slower	Fast	Fast

Types of Mobile Ads

1-Mobile Video Advertising

Video advertising is a fast-growing industry because of its high conversion rates when promoting a product or service

2-Banner Advertising

- A banner advertisement is a thin horizontal graphic that stretches across a screen to promote a brand or product.

3-Pop-ups

- Pop-ups are windows that appear on your screen and overlay the host website's page. These are great for improving conversion rates and building email lists.

4-Native Advertisements

- Native advertisements are built to blend into the environment or app that they are displayed on. A significant advantage of native advertising is that ads can't be exited or blocked by the user.

5-Interstitial Advertisements

- Interstitial advertisements are used at transition points within apps and take up the full screen of your mobile device. Some examples of transition points occur while a page is loading, in between game levels, or when opening an app.

6-Location Advertising

- Location advertising (also known as geotargeting) is the process of sending targeted messaging to individuals that enter a specific location. Geotargeting is a great method for gathering potential customer data by seeing which stores they frequent and tapping into their interests.

What Is a Call to Action (CTA)?

- A call to action (CTA) is a marketing term that refers to the next step a marketer wants its audience or reader to take.

الخطوة التالية بعد مشاهدة المحتوى

- For example, it can instruct the reader to click the buy button to complete a sale, or it can simply move the audience further along towards becoming a consumer of that company's goods or services.

Types of Gaming Business models :

1. **Game purchase (Premium games):** where the digital gaming company sells it at a given price. Sells at a given price
2. **In-app purchases (Freemium games):** similar to the free-to-play model, the free game prompts gamers to buy either digital goods or premium versions/upgrades of the game. Buy digital goods/upgrades
3. **Ad-supported:** another model is the ad-supported, where gamers get it for free, and advertising is rendered within the game, which will monetize on both an impression and click basis.
4. **Hybrid model games:** A mix between In-app purchases and Ad-display which lets you pay to get rid of the advertisements.
5. **non-game focused monetization** : like videos about game playing Videos about playing

Game Testing Checklist

10 -Performance

1. **Loading** time: Check the loading time of the game while starting the app for the first time
2. **Smooth** Loading: Check if the loading continues smoothly without any interruptions loading without interruptions.
3. **Speed**: Make sure that no aspect of the game takes too much time to complete actions don't take too much time
4. **Flow**: Verify if the game hangs or gets stuck in-between and analyze the reasons behind it verify if game hangs or gets stuck
5. **Check** the device memory size of the device during idle and running state of the game
6. **Multitasking** - Test if the game app works fine while switching between applications.

Game Testing Checklist

10 -Performance-2

7. **Memory Leak** - Battery low information must appear on screen.
8. **Device OS** - Check for platform independence, whether the game works as expected on respective platforms.
9. **Interruption** - If the game is active and the user receives any call, or wants to open bluetooth during that time, then the game must retain the position where it was left. Retain position after interruption.
10. **Upgrade Game** - Game must upgrade itself while keeping the data(score, user profiles) of the user intact.
11. **Battery Effect** - Check what happens if suddenly the battery goes down and the phone gets switched off, the scores are saved or not.
Scores saved if phone switches off.

صل علي سيدنا محمد

ادعوا لي بالتوفيق ولكم بالمثل

افتكروا ابي و جدتي في دعائكم